

# Outline

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- Primary Memory - ROM

# Primary Memory - ROM

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## **ROM (Non-volatile Memory)**

- It is a non-volatile memory.
- It means it does not lose its data or programs that are written on it at the time of manufacture.
- So it is a permanent memory that contains all important data and instructions needed to perform important tasks like the boot process.



# Primary Memory – Types of ROM

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## **Programmable Read Only Memory (PROM) :**

- is a blank version of ROM.
- It is manufactured as blank memory and programmed after manufacturing.
- We can say that it is kept blank at the time of manufacturing.
- You can purchase and then program it once using a special tool called a programmer.

# Primary Memory – Types of ROM

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## **Erasable and Programmable Read Only Memory (EPROM):**

- EPROM is a type of ROM that can be reprogrammed and erased many times.
- The method to erase the data is very different; it comes with a quartz window through which a specific frequency of ultraviolet light is passed for around 40 minutes to erase the data.
- So, it retains its content until it is exposed to the ultraviolet light.
- You need a special device called a PROM programmer or PROM burner to reprogram the EPROM.



# Primary Memory – Types of ROM

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## Electrically Erasable and Programmable Read Only Memory (EEPROM):

- EEPROM is a type of read only memory that can be erased and reprogrammed repeatedly, up to 10000 times.
- It is also known as Flash EEPROM as it is similar to flash memory.
- It is erased and reprogrammed electrically without using ultraviolet light.
- Access time is between 45 and 200 nanoseconds.

# Difference between RAM and ROM

| RAM                                                                      | ROM                                                                                 |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| It is a Random-Access Memory.                                            | It is a Read Only Memory.                                                           |
| Read and write operations can be performed.                              | Only Read operation can be performed.                                               |
| Data can be lost in volatile memory when the power supply is turned off. | Data cannot be lost in non-volatile memory when the power supply is turned off.     |
| It is a faster and expensive memory.                                     | It is a slower and less expensive memory.                                           |
| Storage data requires to be refreshed in RAM.                            | Storage data does not need to be refreshed in ROM.                                  |
| The size of the chip is bigger than the ROM chip to store the data.      | The size of the chip is smaller than the RAM chip to store the same amount of data. |
| Types of RAM: DRAM and SRAM                                              | Types of ROM: MROM, PROM, EPROM, EEPROM                                             |



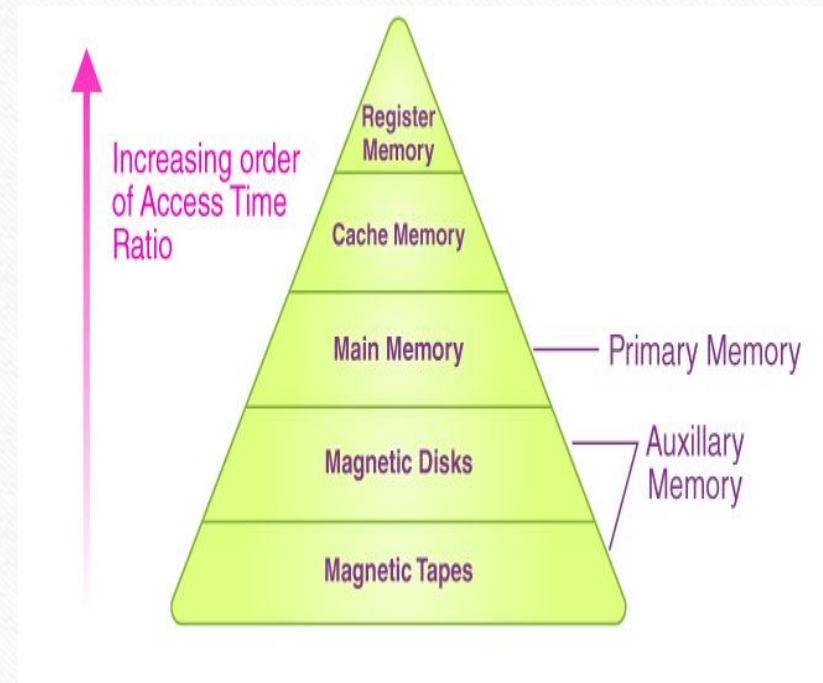
# Memory Units/Memory Capacity

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- 1) **Bit:** The computer memory units start from bit. A bit is the smallest memory unit to measure data stored in main memory and storage devices. A bit can have only one binary value out of 0 and 1.
- 2) **Byte:** It is the fundamental unit to measure data. It contains 8 bits or is equal to 8 bits. Thus a byte can represent  $2^8$  or 256 values.
- 3) **Kilobyte:** A kilobyte contains 1024 bytes.
- 4) **Megabyte:** A megabyte contains 1024 kilobytes.
- 5) **Gigabyte:** A gigabyte contains 1024 megabyte.
- 6) **Terabyte:** A terabyte contains 1024 gigabytes.

# Memory Hierarchy

- The computer memory can be divided into 5 major hierarchies that are based on use as well as speed.
- A processor can easily move from any one level to some other on the basis of its requirements.
- These five hierarchies in a system's memory are register, cache memory, main memory, magnetic disc, and magnetic tape.





# Properties or Characteristics of a Memory Device

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## **Access Time :**

- Average time required to read a fixed amount of information from the memory.

## **Access Mode :**

- It defines the manner in which the records of the file are made available for processing.

Two type of access mode :

- Serial Access Mode
- Random Access Mode

## **Serial Access Mode :**

- Serial access, involves accessing data sequentially, one piece at a time, typically starting from the beginning of the data and moving in order to the desired location.
- To access data in a serial access device, you must read or write the data sequentially, which means you need to go through the data that comes before it.

# Properties or Characteristics of a Memory Device cont..

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## **Random Access Mode :**

- Random access refers to the ability to access data at any location within a storage device directly, without having to read through the data sequentially.
- With random access, you can jump to any specific data location within the storage device without going through the preceding data.

## **Cycle Time :**

- The minimum time between two consecutive memory access operation is called the cycle time.

## **Data Transfer Rate :**

- The maximum amount of information that can be transferred to or from the memory every second is called the data transfer rate.



# Thank You

